

Final Year Project Seminar Series



REALISTIC CONSTRAINTS

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Presentation Outline

- ◉ Definition
- ◉ Criteria and Constraints
- ◉ Problem Characteristics
- ◉ Guideline Lists of Realistic Constraints
- ◉ Social Example
- ◉ Engineering Example
- ◉ Conclusion

Definition

Realistic Constraints:

A constraint is a design decision imposed by the environment or a stakeholder that impacts or limits the design being built. Some of the realistic constraints include: performance, economical, energy, environmental, sustainability, manufacturability, maintainability, operational, reliability and availability, usability, social and cultural, legal, political, ethical, health and safety, security, interoperability, portability, reusability, robustness, scalability, testability, traceability, user-friendliness (usability), conformity to standards, *etc.*

Definition

In addition to the traditional technological and economic considerations fundamental to the design of mechanical components and systems, the modern engineer has to be increasingly concerned with the broader considerations of realistic constraints which are particularly related to the better-off today's society and quality of life. This task challenges the engineers to assemble all pertinent facts, and then to make good decisions through better understanding, imagination, ingenuity, and judgment.

For your FYP design, you must be imaginative and creative enough to predict potentially hazardous situations and all the factors relating to the component /system and make the best design decision to address those **realistic constraint issues**. You also need to check **government and industry codes and standards and the pertinent technical literature to be sure that legal requirements** are complied with.

Criteria and Constraints

An important step in design: Establish design criteria and constraints.

Design criteria are requirements you specify that will be used to make decisions about how to build and evaluate the product. Criteria are derived from needs expressed by customers. Criteria define the product physical and functional characteristics. **Some examples of criteria are shape, size, weight, speed, ruggedness, and ease of manufacture.**

Constraints are factors that limit the engineer's flexibility. Some typical constraints are **cost, time, and knowledge; legal issues; natural factors such as topography, climate, raw materials; and where the product will be used.** Good designs will meet important design criteria within the limits fixed by the constraints. Good designs are also economical to make and use because **cost is always a design constraint!**

Problem Characteristics

➤ Engineering Problem

- Problem statement incomplete, ambiguous
- No readily identifiable closure
- Solutions neither unique nor compact
- Solution needs integration of many specialties

➤ Science Problem

- Concise problem statement
- Identifiable closure
- Unique solution
- Problem defined and solved with specialized knowledge

Guideline Lists of Realistic Constraints

Performance: The performance demanded or likely to be demanded should be fully defined. Examples of items to be considered include: how often the device will be used; likely loading patterns; maximum RAM/memory etc.

Economical: This requirement addresses the cost associated with the development. The goals for an engineering design, related to the economical constraints, are to minimize the materials and energy for products/production, engineering effort, and time to market, as well as to maximize the value to consumers, and efficiency of production. Pertinent questions include:

- 1) What are the development, production and operational/maintenance costs?
- 2) Prices of current related or similar products in market.
- 3) Potential impact to the local and regional economy
- 4) Budget limitations *e.g. The total cost the project should not exceed 200 RO.*

Guideline Lists of Realistic Constraints

Environmental: This requirement addresses the impact of the component / system on the external environment and usage of earth's resources. The goals related to the environmental constraints, are to use **recyclable materials** in manufacturing and production wherever possible, **clean production**, use **alternate materials** or approaches, **minimize energy** use in manufacturing, production and shipping, **minimize waste** and **prevent pollution**. Pertinent questions include:

- 1) What valuable resources are being used? (Palm tree leaves)
- 2) Is pollution being produced as a result of production and/or use of the product?

Energy: Constrains on the amount of energy the component / system consumes.

Guideline Lists of Realistic Constraints

Social and Cultural: address constraints that have to be perceived by the intended user or by society at large (video games; children toys, amusement parks); benefits, risks, and acceptance of products. You must be aware of the cultural sensitivities as well as the values and believes of the countries in which the product will be deployed and used. For example, if the requirements conflict in other countries or societies, several versions of the product might have to be developed and marketed. An example constraint could be that the product shall provide help menus to the user in either English or Arabic.

What anticipated impacts will the product have within a community?

e.g. Does this product offend a minority?

e.g. How much violence should be in this video game?

e.g. What are the government laws to protect society?

Guideline Lists of Realistic Constraints

Political: Political constraints address relationships to political, governmental, or union organizations. Sample questions to be answered:

- Is the project being conducted for a governmental entity?
- Is the project being regulated, or being proposed for regulation, by a governmental entity?
 - a) Products (e.g., computer games, marks on clothes) that profile negative sides of a specific race or gender
 - b) Products for customers who are against / for certain countries (cars)

Conformity to standards: Standards conformity requirements indicate the standards that must be followed while designing the component / system. There are different levels of standards. Internal (within company), national (Oman), regional (GCC, EU), international (ISO), professional (ASME, IIE, ASTM, etc.), etc.

Guideline Lists of Realistic Constraints

Ethical: Ethical constraints address issues of professional responsibility such as privacy, competence, intellectual property rights, misuse, etc. The goals for an engineering design, related to the ethical constraints, are to avoid conflicts of interests, reject bribery, be honest with claims, estimates, research and testing, maintaining and improve technical competence, seek and offer honest criticism, correct errors, credit contributions of others, treat others fairly, assist colleagues and coworkers, work honestly, and deliver good engineering value to the customer and end user.

Are there any foreseen potential conflicts with a profession's Code of Ethics arising from the development or use of the product? Designs without considering safety and health of workers, consumers, and/or the public? Products implicitly using patent protected designs/concepts? Under design for profit? Use of potentially health hazard materials.

Guideline Lists of Realistic Constraints

Safety: Understand any safety aspects, safety standards, and legislation covering the product type. This includes the need for labeling, safety warnings, etc. Consider various safety aspects relating to mechanical, chemical, electrical, thermal, etc.

Accuracy and Reliability: Establish limits for precision (repeatability) and accuracy (how close to the "true" value) and the range over which this is true of the device. [\(Machines in Engineering Workshop\)](#)

Shelf Life: Establish environmental conditions while in storage, shelf life of components such as [Juice Bottles, Solar Panels, Batteries](#), etc.

Aesthetics, Appearance, and Finish: Color, shape, form, and texture of finish should be specified where possible ([take opinions](#)).

Guideline Lists of Design Criteria

Ergonomics: Establish restrictions on the interaction of the product with human, including heights, reach, acceptable forces / torques, etc.

Size: Establish restrictions on the size of the product, including maximum size, portability, space available, access for maintenance, etc.

Weight: Establish restrictions on maximum, minimum, and/or optimum weight; weight is important when it comes to handling the product by the user, by the distributor, handling on the shop floor, during installation, etc.

Materials: Establish restrictions if certain materials should be used and if certain materials should NOT be used (**ferrous materials in MRI machine**).

Operating Environment: Establish the conditions that the device could be exposed to during operation (or at any other time), including **temperature range**, pressure range, **humidity**, shock loading, **dirt or dust**, **corrosion from fluids**, noise levels, insects, vibration, etc.

Guideline Lists of Criteria / Realistic Constraints

Manufacturability:

- 1) Designs with an impossibly small manufacturing tolerance e.g. $\pm 0.003\text{mm}$
- 2) Designs with a required highly accurate first natural frequency
- 3) Designs with an impossibly high stiffness
- 4) Designs with a zero-friction contacting surface
- 5) Can the proposed material be welded if welding is the proposed assembly method?
- 6) Is the product's surface paintable if it is designed to have an artificial color?
- 7) Titanium alloy, tool steel and ceramics require special cutting tools (**Not available in our workshops; no technical expertise**)

K N O W L E D G E

Social Example

Situation Description: An engineer must consider realistic constraints such as economical, safety, reliability, aesthetic, social impact, technical and feasibility factors in his design. Assume that you want to buy a gift to "your 5-year old brother" for the Eid. What are the feasible alternatives? Which one appeals to you better and how you apply the realistic design constraints to your selection?

Conditions: a) The budget that you have for the toy is 10 RO; b) The toy must be safe to play; c) It must be durable to harsh use by the kids; d) It must contribute to the mental and social development of the kids; e) Educate the kid in matching shapes, constructing useful items from components etc.; f) Suitable to play by a group of kids; g) It must not cause disturbance to surrounding during the play; h) It must have a minimal running cost.; i) Available in the local market.

Social Example

Choices within the budget range:

- a) A pocket knife
- b) Color pens and a paint book
- c) Puzzle
- d) A remote controlled car
- e) A mobile toy transceiver set
- f) A model aircraft
- g) A model robot
- h) A LEGO kit

Out of 8 selections, "f" and "g" are not available locally. Choices "d" and "e" require batteries to operate that contribute to the maintenance cost. Choice "a" is not safe. Among the remaining three choices, the **LEGO kit** appeals better in satisfying the condition "d" and it is the final choice.

Engineering Examples

Omani Bread Making Machine:

Designed by one group of mechanical engineering students and later refined to make working machine by the second group. However, realistic constraints were missing e.g.

- The total cost was 400 RO.
- No safety aspect: hot surfaces are exposed
- Handling: putting the dough and taking out the bread were left to the user
- Messy after cooking the bread
- Half of the design

Engineering Examples

1st, 2nd and 3rd Generations of SQU Solar Car:

Designed by three groups of mechanical engineering students, followed by other groups over few years. What were the realistic constraints e.g.

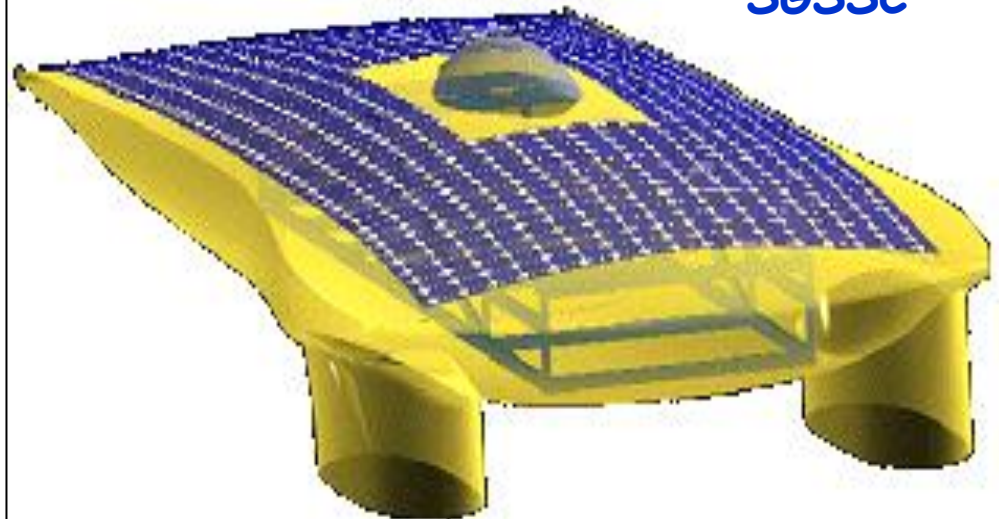
- **Economic:** Budget limitations (95,000 \$ v/s over 1000,000 \$)
- **Social:** restrictions on generating fund ; no failure
- **Legal / Rules:** Surface area < 10 m²; width < 2 m; mass < 300 kg
Any maintenance work will result in deduction of points
- **Manufacturability:** No casting facility in SQU and surrounding
& **Material** Restricted availability of materials/sizes in GCC
- **Internal:** How to design with less # of batteries and run at 60 KPH?
Easy to modify
- **Standard:** Driver safety

Body Design

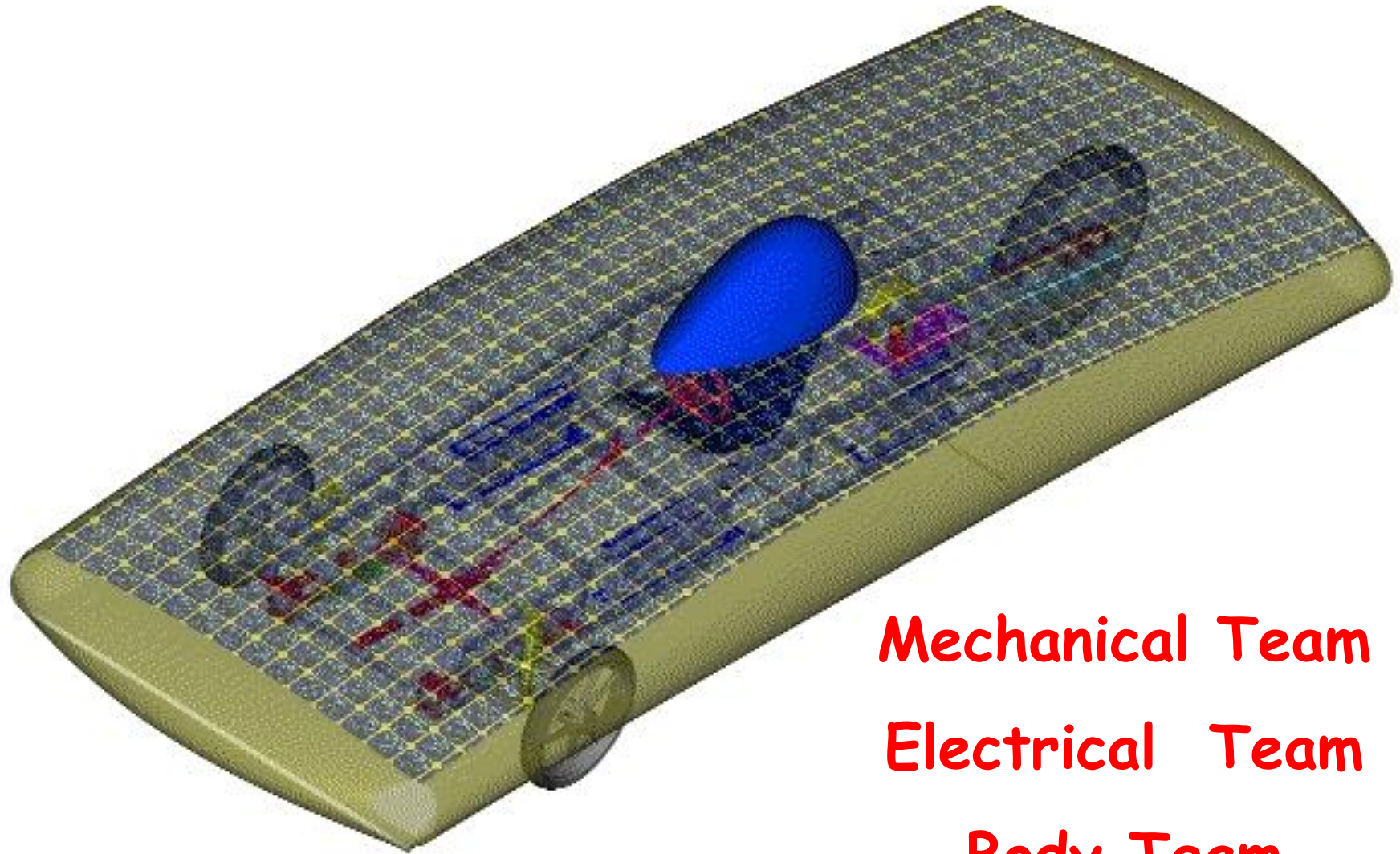
FGSSC



SGSSC



Team Divisions of the Second Generation SQU Solar Car Project



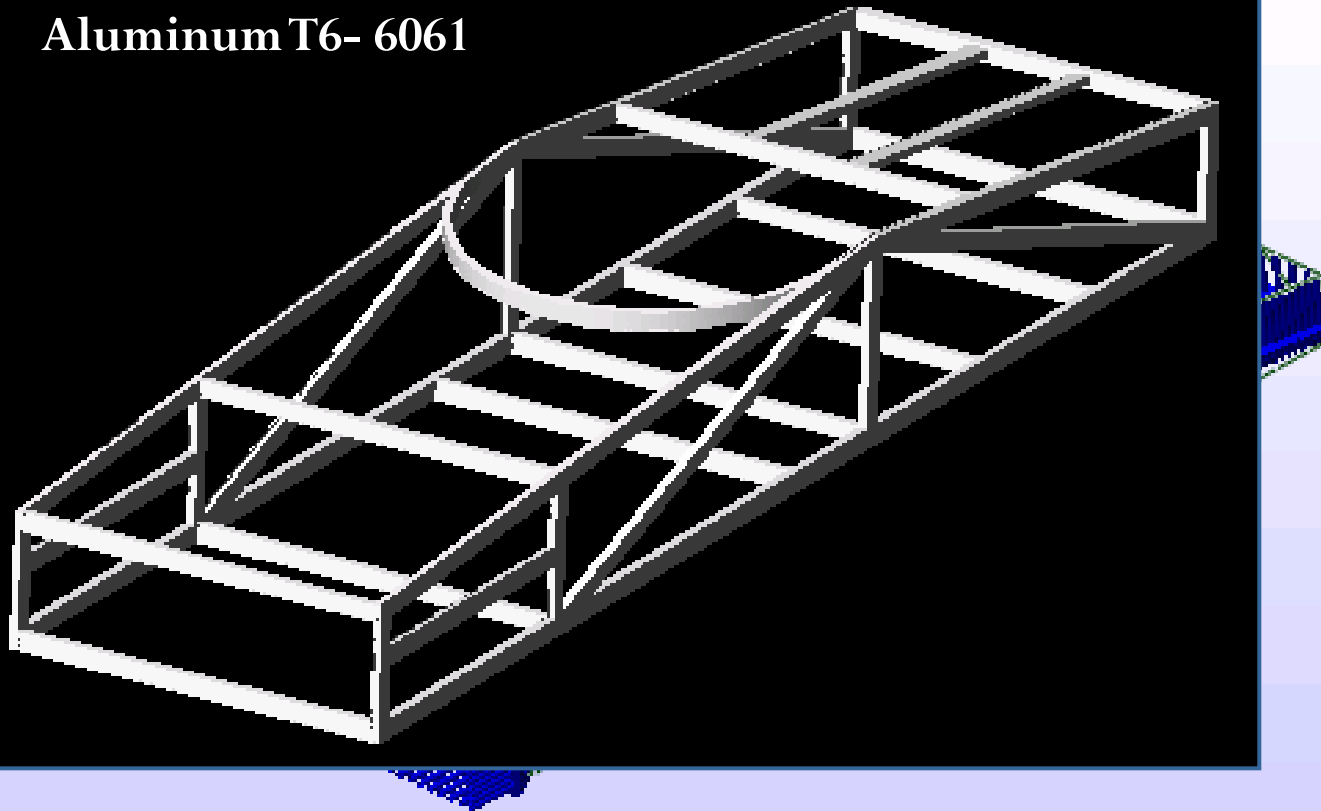
Mechanical Team

Electrical Team

Body Team

Chassis

Aluminum T6- 6061



ss

e+008

e+008

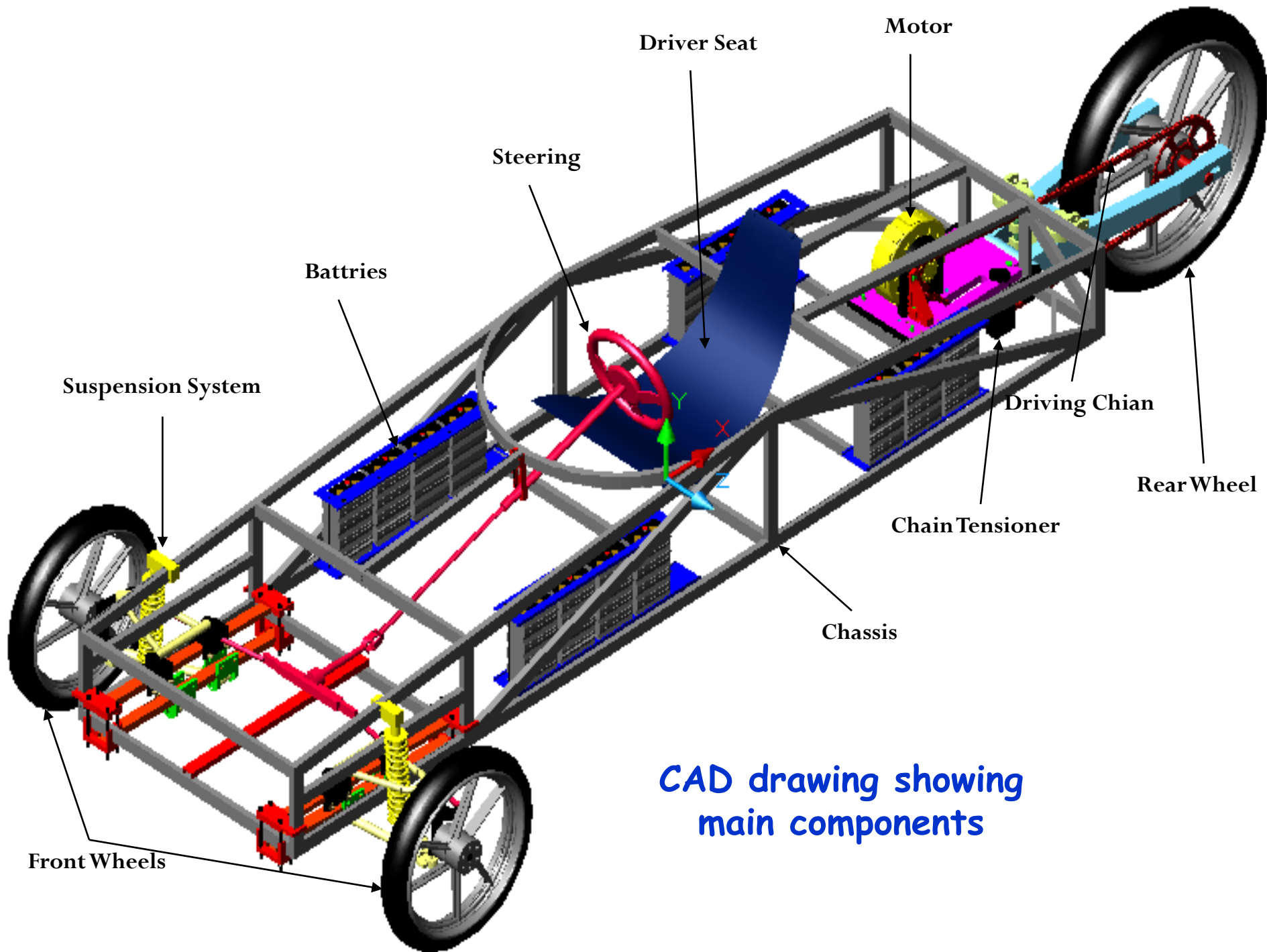
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Fabrication Process



Fabrication Process



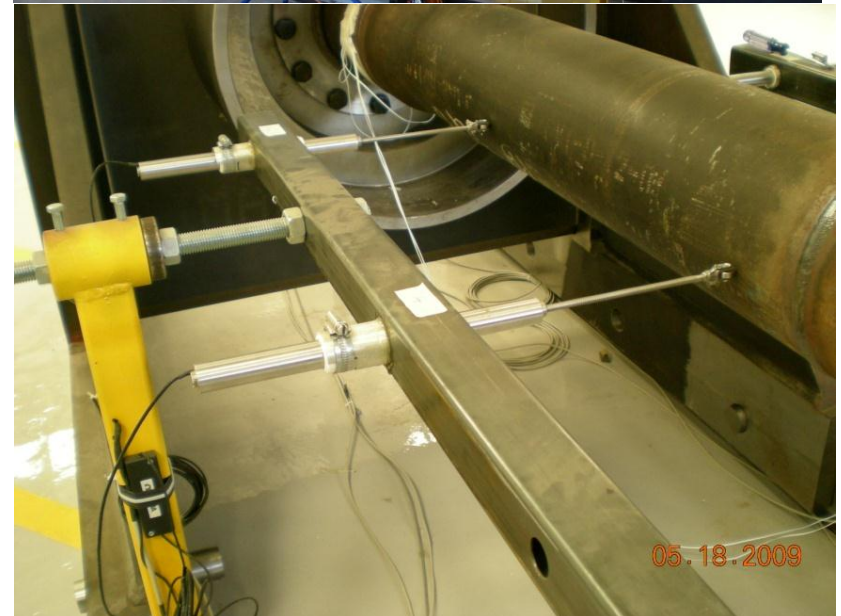
Engineering Examples

Expandable Tubular Test-Rig:

- 1) Has to serve global community v/s local only (**variable diameter v/s two diameters 7.5/8" and 9.5/8"**)
- 2) HSE requirements particularly safety (**high strength hoses, trenches & gratings, alarms, bullet proof glass, automatic shut off,**)
- 3) Safe against burst and collapse
- 4) Legal: **intellectual property violations, confidentiality,**
- 5) High strength: sustain a maximum load of 250 tons with a safety factor of 1.4 (**materials with $S_y = 1200$ MPa, welding**)
- 6) Easy to assemble and disassemble with manual/semi automatic tools and local labor (**use of sensors, DTOP, etc.**)

Expandable Tubular Test-rig





Delfly

Constraints on the design:

- Weight: close to 15 g
- Wingspan: less than 45 cm
- Cost: within a budget of € 5000
- Noise level: less than 60 dB at a distance of 15 m
- Endurance: at least 5 min
- Slow flight: less than 5 m/s
- Use of COTS components



Converter from 240 V to 120 V

The design of the portable converter makes it compact and with clean lines. However, it was never tested in European sockets. The prongs of the plug are not long enough to reach into the recessed sockets that are now the norm.



Conclusion

- 1) The crux of the design process is in creating a satisfactory solution to a need.
- 2) Beware of the difference between criteria and constraints.
- 3) Clearly list the criteria and constraints for your design exercise and keep reviewing it till reach your goal.
- 4) Realistic constraints provide the boundary conditions necessary for the design decision.
- 5) For FYP it is important and necessary to focus on undergraduate design than on undergraduate research

***THANK YOU FOR
YOUR ATTENTION &
PRESENCE***